content. The p-value $\widehat{p}'$ quantifies the strength of evidence against the null hypothesis.

P-Value Computation The p-values for assessing the shifts in accuracy are computed using the standard normal cumulative distribution function Φ as follows:

$$\widehat{p} = 1 - \Phi\left(\frac{acc_1 - acc_2}{\sqrt{\frac{acc_1(1-acc_1)}{n_1} + \frac{acc_2(1-acc_2)}{n_2}}}\right), \quad (6)$$

acc_1 and acc_1 could be either acc_{past} or acc_{present}, which is based on which Hypothesis Testing is used. Note that the two Hypothesis Testings are symmetrical.

Detailed experiment settings are listed in Appendix D.

4.1.2 Observation

Models tend to be stronger in earlier periods before release, with performance gradually declining over time. Using the -20 to 0 months period as an anchor, if we assign the three periods to represent the near past (20 to 40 months before release), the middle past (40 to 60 months before release), and the distant past (60 to 80 months before release), we observe that the further back in time we go, the more pronounced the Nostalgia Bias becomes, and it is also more significant. This suggests that the models gradually degrade during the pre-release period, rather than experiencing a rapid decline around the cutoff period.

Nostalgia Bias is prevalent. As shown in Tab. 2, among the biases, Nostalgia is the most common, followed by periods where neither bias is significant (shown as -), with Neophilia Bias being the least frequent. Phi-1.5 and Baichuan2-13B series show Nostalgia Bias in all three periods before release. Mistral-7B-v0.1 show Neophilia Bias in 20 to 40 months before release but show more significant Nostalgia in another two periods, which we attribute to random fluctuations rather than a consistent trend.

This observation reveals that models generally favor older knowledge and prefer information from more distant past periods. The distribution of knowledge is not uniform, nor does it tend to be closer to the present. This poses a potential risk in our utilization of these models because there is an expectation that the knowledge utilized should be as up-to-date as possible, ideally close to the model's release date.

4.2 Temporal Degeneration

4.2.1 Protocol

We discussed generally *Temporal Degeneration* in 2.1.2, here we need to define specifically.

Definition 4.1. Temporal Degeneration The capability of future event prediction might be degraded over time due to the larger generalization gap brought by time.

To qualitatively measure Temporal Degeneration, we propose a hypothesis testing like 4.1.1.

Hypothesis Testing 3. Temporal Degeneration

$$\begin{aligned} T_3: \quad & H_0: \mathbf{E}[acc_{\text{future}}] = \mathbf{E}[acc_{\text{present}}], \\ & H_1: \mathbf{E}[acc_{\text{future}}] < \mathbf{E}[acc_{\text{present}}] \end{aligned} \quad (7)$$

Similar to Sec 4.1.1, acc_{future} represents the model accuracy on questions proposed after the model's release. This hypothesis assesses whether there is a significant decline in model accuracy, indicating a degeneration over time. The hypothetical p-value is similar to that of Eq. 6.

4.2.2 Observation

Earlier models perform worse, while newer advanced models perform better in future predictions. As the latest scores in Table 3 show, earlier models, show a significant performance decline across multiple time periods. For example, LLaMA-7B's accuracy dropped from 0.11 in March 2023 to 0.08 by March 2024, with no improvement in later evaluation periods. In contrast, newer models exhibit much stronger future prediction capabilities.

Close-source models perform well initially but struggle with temporal generalization, while open-source models show better stability. As is shown in Appendix C, in both pre- and post-release scenarios, proprietary models such as GPT-4 and Claude-3.5 slightly outperform their open-source counterparts. However, when predicting outcomes for events known only post-release, all models exhibited substantial declines in performance.

Our analysis highlights models with less than 31% decline in post-release accuracy in **bold**, illustrating relatively stable performance. Conversely, models that experienced declines greater than 39% are marked by <u>underline</u>, indicating substantial performance degeneration. Notably, models such as Command R+, Mixtral-8x22B-Instruct-v0.1, DeepSeek-V2-Chat, and LLaMA-3.1-405B-Instruct demonstrate slower declines in accuracy

Table 2: Comparison of Model Performance in Different Pre-Release Periods (Compared to 0-20 months before Release), uses asterisks to denote significance levels:† ($p < 0.05$), ††($p < 0.01$), and ††† ($p < 0.001$), → refers to Neophilia Bias, ← refers to Nostalgia Bias.

Models	20-40 Months Before Release	40-60 Months Before Release	60-80 Months Before Release
OPT-13B	–	← †††	← ††
OPT-2.7B	–	→ †	–
LLaMA-7B	–	← †††	← ††
Pythia-12B	→ ††	← †††	← †††
Falcon-rw-1B	–	–	–
Baichuan-7B-Chat	–	–	← †
LLaMA-2-13B	–	← †††	← †††
LLaMA-2-7B	← †††	← †	–
LLaMA-2-7B-Chat	–	← †††	← †††
Zhongjing-Base	–	–	← †††
InternLM-Chat-7B	← ††	–	–
Baichuan2-7B-Chat	← ††	–	–
Mistral-7B-v0.1	→ †	← ††	← †††
Phi-1.5	← †	← ††	← †††
Baichuan2-13B-Base	← ††	← †††	← †††
Baichuan2-13B-Chat	← †	← †††	← †††
Colossal-LLaMA-2-7B-Base	← ††	–	← †
Qwen-14B-Chat	← ††	← ††	← †††
Qwen-7B	–	–	–
Skywork-13B-Base	–	–	–
Zephyr-7B-beta	← †	← †††	← †††
Yi-6B	–	–	–
Yi-6B-Chat	← †	← †	← †
Qwen-1.8B	–	–	← ††
RWKV-v5-Eagle-7B	–	–	–
Phi-2	← ††	–	← †
Command R+	← ††	← †††	← †††
Mixtral-8x22B-Instruct-v0.1	–	← †††	← †††
Phi-3-mini-4k-instruct	–	← †††	← ††
Qwen1.5-110B-Chat	← †	← †††	← †††

post-release. Conversely, models like Claude-3.5-Sonnet-20240620, GPT-4-231106, and GPT-4o show rapid declines in performance,

The case studies of model iterations within the Claude series and the GPT models illustrate the dynamics of model evolution and generalization capabilities over time. Notably, the Claude-3-opus-20240229 model shows the least decline, indicating a potentially better tuning for broader applications or general datasets. This contrasts with the more significant declines seen in Claude-3-sonnet-20240229 and Claude-3.5-Sonnet-20240620, suggesting that these iterations may have undergone specific architectural tweaks, utilized different training data scopes, or implemented optimization strategies that did not generalize as effectively.

Similarly, the progression from GPT-3.5-turbo-230613 to GPT-4-231106 shows an increasing trend in both pre-release accuracy and the magnitude of decline post-release, with the GPT-4-231106 model experiencing the largest drop (39.39%). This trend could indicate a push towards optimizing these models for higher initial accuracy, possibly at the expense of their ability to generalize effectively over time.

Miniature versions of models show significant declines in temporal performance. The data reveals that miniature versions of models underperform compared to their full-sized counterparts in temporal tasks. For instance, Gemini.pro had an accuracy of 0.37 in March 2024, but this dropped sharply to 0.17 by July 2024, indicating a significant loss in precision for temporal tasks. Similarly, the mini version of GPT-4 showed considerably lower performance across multiple time periods, particularly in April 2024, where its accuracy was just 0.21, around 0.1 lower than the full-sized GPT-4 in the same period. This illustrates that while mini models improve inference efficiency, they exhibit poor generalization over time, with perfor-

Table 3: Accuracy performance of various models across different periods. The presence of asterisks (*) denotes statistically significant decreases in performance during specific periods (p < 0.05 for *, p < 0.01 for **, and p < 0.001 for ***), benchmarked against their accuracy before release. Grey cells indicate the models have not been released in this period and represent the average scores of the models prior to their release.

Model	23/1 -23/2	23/3 -23/4	23/5 -23/6	23/7 -23/8	23/9 -23/10	23/11 -23/12	24/1 -24/2	24/3 -24/4	24/5 -24/6	24/7 -24/8
Baichuan-13B-Chat	0.52		0.26**	0.20**	0.36**	0.32**	0.33**	0.20**	0.23**	0.17**
GPT-3.5-turbo-230613	0.49		0.29**	0.37*	0.45	0.44	0.37*	0.29**	0.28**	0.26**
GPT-4-230613	0.6		0.39**	0.43*	0.41**	0.38**	0.48	0.40**	0.35**	0.36**
Llama-2-13B	0.52			0.43	0.48	0.36**	0.39*	0.40*	0.28**	0.34**
Llama-2-7B	0.27			0.26	0.27	0.22	0.26	0.33	0.18**	0.24
LLaMA2-7B-Chat	0.45			0.30*	0.46	0.35**	0.37	0.38	0.27**	0.28**
Baichuan2-13B-Base	0.49			0.30**	0.46	0.42*	0.28**	0.39*	0.27**	0.27**
Baichuan2-13B-Chat	0.5			0.33**	0.43	0.41*	0.26**	0.29**	0.29**	0.33**
Baichuan2-7B-Base	0.42			0.26**	0.30*	0.24**	0.33	0.33*	0.25**	0.22**
Baichuan2-7B-Chat	0.37			0.35	0.43	0.31	0.37	0.40	0.23**	0.27*
Colossal-LLaMA-2-7B-Base	0.42				0.43	0.37	0.28*	0.33*	0.25**	0.26**
Mistral-7B-v0.1	0.44				0.41	0.35*	0.28*	0.34*	0.25**	0.24**
Phi-1.5	0.41				0.38	0.38	0.30	0.26**	0.32*	0.27**
Qwen-14B-Chat	0.39				0.25**	0.24**	0.33	0.27**	0.28**	0.19**
Zephyr-7B-beta	0.4				0.32	0.32*	0.28*	0.28**	0.31*	0.19**
Yi-6B	0.29				0.27	0.16**	0.22	0.33	0.21*	0.21*
Gemini	0.43					0.20**	0.37	0.24**	0.21**	0.17**
Qwen-1.8B	0.35					0.22**	0.37	0.29	0.20**	0.20**
GPT-4-231106	0.66					0.45**	0.39**	0.36**	0.47**	0.43**
Phi-2	0.31					0.23*	0.22	0.29	0.18**	0.15**
Claude-3-opus-20240229	0.53						0.52	0.37**	0.47	0.36**
Claude-3-sonnet-20240229	0.23						0.24	0.13**	0.10**	0.07**
Command R+	0.54							0.37**	0.43*	0.33**
DeepSeek-V2-Chat	0.58							0.49**	0.46**	0.35**
Mixtral-8x22B-Instruct-v0.1	0.56							0.39**	0.43**	0.35**
Phi-3-mini-4k-Instruct	0.38							0.26**	0.24**	0.26**
Qwen1.5-110B-Chat	0.56							0.40**	0.45*	0.36**
Claude-3.5-Sonnet-20240620	0.63								0.48**	0.23**
Gemini-1.5-Pro	0.5								0.47	0.35**
Gemini-1.5-flash	0.39								0.23**	0.22*
Qwen2-72B-Instruct	0.6								0.45**	0.40**
LLaMA-3.1-405B-Instruct	0.5									0.41
GPT-4o	0.62									0.39**
GPT-4o-mini-2024-07-18	0.48									0.28**

mance rapidly declining in long-term prediction tasks.

5 Conclusion and Community Call to Action

We rigorously define and quantify temporal generalization and bias within LLMs, uncovering significant challenges in predicting and adapting to future contexts. Our experiments reveal that LLMs often struggle with temporal biases, affecting reliability in dynamic scenarios requiring accurate forecasting. The evaluation framework and datasets we introduce allow for an in-depth assessment of model performance over time. The findings highlight shortcomings in existing benchmarks related to temporal shifts and underscore the urgent need for improvements in LLM training and updating processes to enhance adaptability and mitigate biases.

Specifically, we urge the community to take note of this phenomenon: models tend to better comprehend earlier events. Cheng et al. (2024) suggests that models are most familiar with knowledge up to the year 2020. However, our findings indicate that the understanding of world events (even in cases where models can remember) lingers around 2015, with a trend towards even earlier years (beyond the time range of our test data). This observation is contrary to our initial hypothesis that models may overfit current events, which is significantly more concerning. It suggests that caution is needed, especially for those hoping to use model judgments directly.